

task-1

Q-Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

1.Check BIND9 Service :

```
(kali@kali)-[~]
$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:named(8)

(kali@kali)-[~]
$
```

2. Start and enable BIND9 Service:

```
(kali@kali)-[~]
$ sudo systemctl start named

(kali@kali)-[~]
$ sudo systemctl enable named
Synchronizing state of named.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable named

(kali@kali)-[~]
$
```

3.Check BIND9 Service Status:

```
(kali@kali)-[~]
$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-04 05:21:15 EDT; 2s ago
     Docs: man:named(8)
  Main PID: 13369 (named)
    Tasks: 14 (limit: 4426)
   Memory: 62.9M (peak: 63.5M)
      CPU: 84ms
   CGroup: /system.slice/named.service
           └─13369 /usr/sbin/named -f -u bind
```

4. Zone configuration:

```

(kali@kali)-[~]
└─$ cat /etc/bind/named.conf.local
//
// Do any local configuration here
//

zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
(kali@kali)-[~]
└─$

```

5. Zone file:

```

(kali@kali)-[~]
└─$ cat /etc/bind/db.musicmania.in
$TTL 604800
@      IN  SOA  ns1.musicmania.in. admin.musicmania.in. (
                        2026080401
                        604800
                        86400
                        2419200
                        604800 )

@      IN  NS   ns1.musicmania.in.

ns1    IN  A    192.168.79.128
@      IN  A    192.168.79.128
www    IN  A    192.168.79.128
mail   IN  A    192.168.79.128
ftp    IN  A    192.168.79.128

```

6.Check BIND Configuration:

```

(kali@kali)-[~]
└─$ sudo named-checkconf

(kali@kali)-[~]
└─$ █

```

7.Check Zone file:

```

(kali@kali)-[~]
$ sudo named-checkzone musicmania.in /etc/bind/db.musicmania.
zone musicmania.in/IN: loaded serial 2026080401
OK

(kali@kali)-[~]
$

```

8. Restart BIND9 Service:

```

(kali@kali)-[~]
$ sudo systemctl restart named

(kali@kali)-[~]
$

```

9. Test DNS Resolution using dig

```

(kali@kali)-[~]
$ dig @127.0.0.1 www.musicmania.in

; <<>> DiG 9.20.26-1-Debian <<>> @127.0.0.1 www.musicmania.in
; (1 server found)
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 3571
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: eac38aabc16b979c010000006a71b32fffb91256103435c6a (good)
;; QUESTION SECTION:
;www.musicmania.in.                IN      A

;; ANSWER SECTION:
www.musicmania.in.        604800 IN      A      192.168.79.128

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Tue Aug 04 05:38:55 EDT 2026
;; MSG SIZE rcvd: 90

```

10. Test DNS Resolution using nslookup

```

(kali@kali)-[~]
$ nslookup www.musicmania.in 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   www.musicmania.in
Address: 192.168.79.128

(kali@kali)-[~]
$

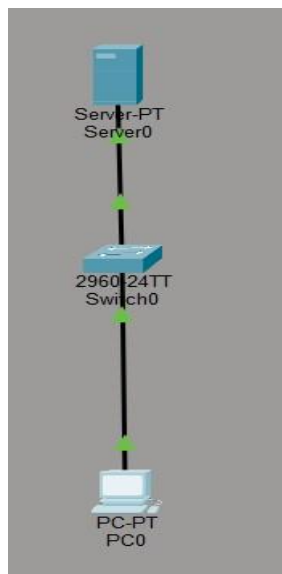
```

Task-2

Q-Add an A record in your DNS server configuration to map `www.musicmania.in` to the IP address `192.168.1.50`, then use the `nslookup` or `dig` command to verify the mapping works.

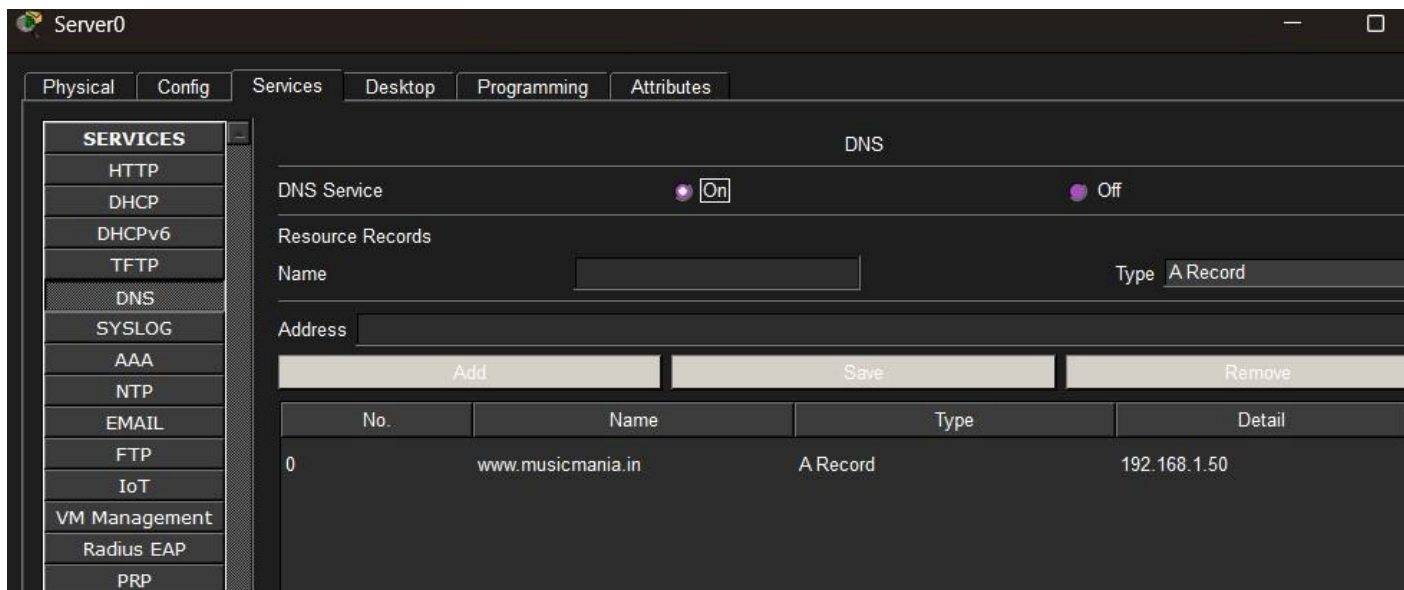
To configure and verify this DNS mapping in Cisco Packet Tracer, must add the record to a Server device configured as a DNS server and query it from a PC.

TOPOLOGY:



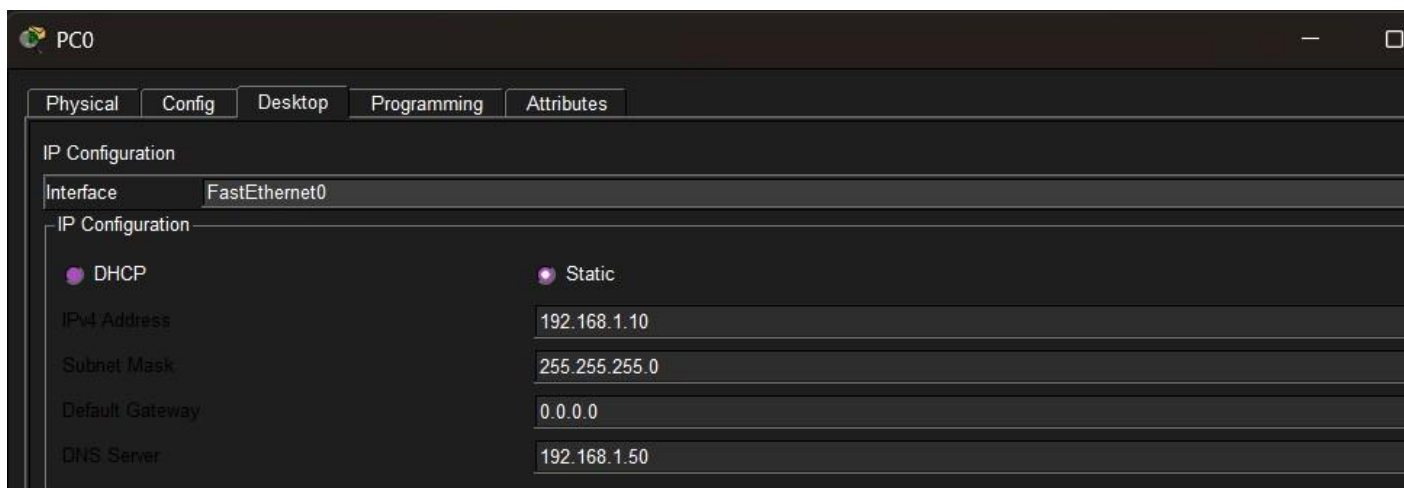
Step 1: Configure the DNS Server

1. Click on the **Server** device in your workspace and open its **Config** or **Services** tab.
2. Select **DNS** from the left-hand services menu. [\[1\]](#) 3. Turn the DNS service **On** by selecting the radio button.
4. Fill in the fields under **Resource Records**:
 - **Name:**
`www.musicmania.in` ○
 - Type:** `A`
Record ○
 - Address:**
`192.168.1.50`
5. Click the **Add** button. The record will appear in the list below.



Step 2: Configure the Client PC

PC: Click PC -> **Desktop** -> **IP Configuration**. Set IP Address to 192.168.1.10, Subnet Mask to 255.255.255.0, and **DNS Server** to 192.168.1.50



TEST OUTPUT:

```
C:\>nslookup www.musicmania.in

Server: [192.168.1.50]
Address: 192.168.1.50

Non-authoritative answer:
Name: www.musicmania.in
Address: 192.168.1.50

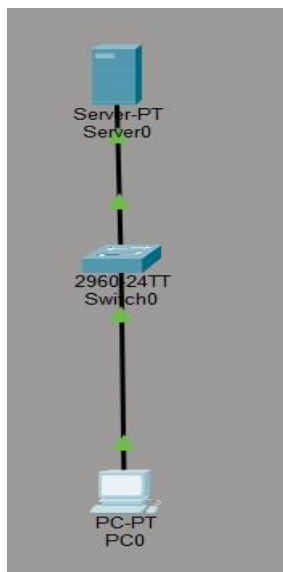
C:\>
```

Task-3

Q-Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

To configure a DNS server by creating an A Record for www.musicmania.in and a CNAME Record for play.musicmania.in, and verify that both domain names resolve to the same IP address using the nslookup command.

TOPOLOGY:



Step 1: Configure the DNS Server

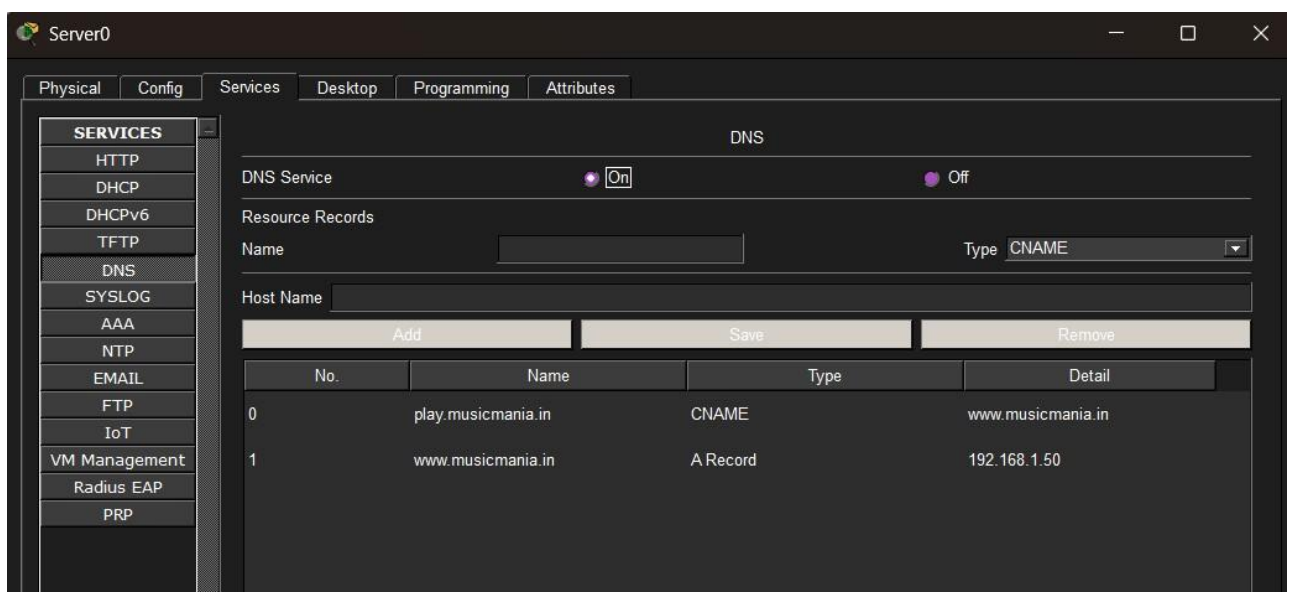
Open **Server0**.
 Select
Services →
DNS. Turn
DNS Service
ON.

- **Name:** www.musicmania.in
- **Type:** A Record
- **Address:** 192.168.1.50 Click **Add**.

Create CNAME Record

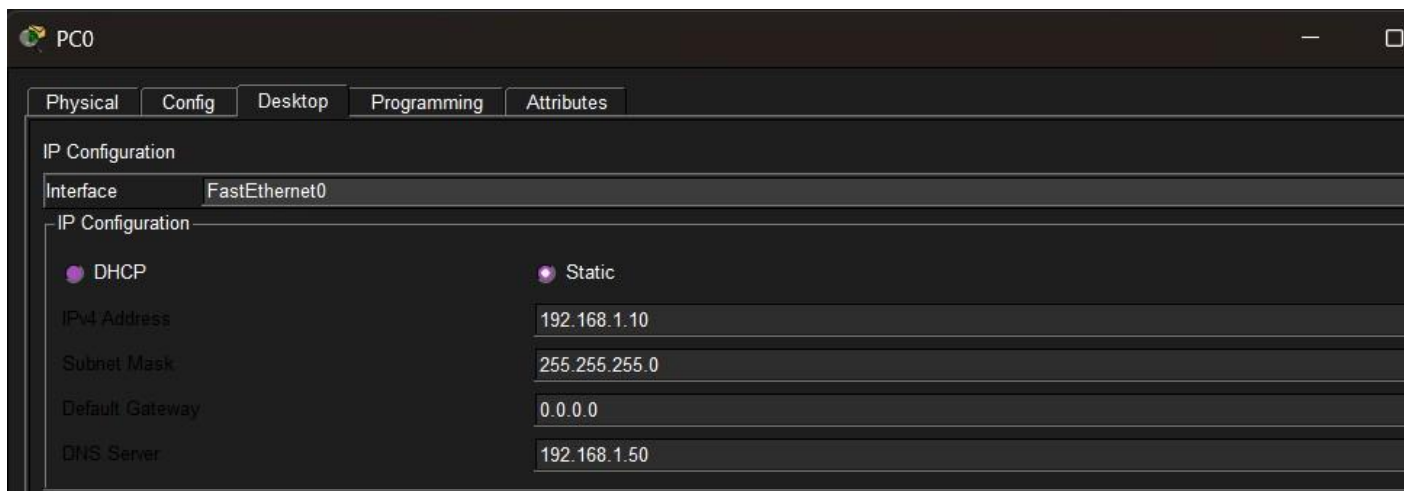
Enter:

- **Name:** play.musicmania.in
- **Type:** CNAME
- **Host Name:** www.musicmania.in Click **Add**.



Step 2: Configure the Client PC

PC: Click PC -> **Desktop** -> **IP Configuration**. Set IP Address to 192.168.1.10, Subnet Mask to 255.255.255.0, and **DNS Server** to 192.168.1.50



TEST OUTPUT:

```
C:\>nslookup play.musicmania.in
Server: [192.168.1.50]
Address: 192.168.1.50

Non-authoritative answer:
Name:   www.musicmania.in
Address: 192.168.1.50

Aliases:   www.musicmania.in
C:\>
```

Task-4

DNS Troubleshooting Process for musicmania.in

Objective

To identify why the DNS server is not resolving musicmania.in correctly and restore proper DNS name resolution.

Step-by-Step Troubleshooting

1. Check Network Connectivity

First, verify that the DNS server has network connectivity.

ping 8.8.8.8

If the ping fails, check the server's network configuration, gateway, and Internet connectivity before troubleshooting DNS.

2. Check DNS Configuration Files

On a Linux DNS server using BIND, check the main configuration files:

```
cat /etc/bind/named.conf
```

```
cat /etc/bind/named.conf.local
```

Check that the musicmania.in zone is configured correctly. For example:

```
zone "musicmania.in" {  
    type master;  
    file "/etc/bind/db.musicmania.in";  
};
```

Then check the zone file:

```
cat /etc/bind/db.musicmania.in
```

Verify that the correct **A record**, nameserver records, and IP address are present.

3. Validate the DNS Configuration

Use:

```
sudo named-checkconf
```

This checks the BIND configuration for syntax errors.

Then check the zone file:

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

If these commands report errors, correct the configuration or zone file before continuing.

4. Restart the DNS Service

After making configuration changes, restart the DNS service:

```
sudo systemctl restart bind9
```

Check whether the service is running:

```
sudo systemctl status bind9
```

If the service does not start, examine the logs:

```
sudo journalctl -u bind9
```

5. Test DNS Resolution Using nslookup

Test the domain against the local DNS server:

```
nslookup musicmania.in
```

The response should contain the expected IP address.

You can also specify the DNS server directly:

```
nslookup musicmania.in 192.168.1.10
```

Replace 192.168.1.10 with the actual DNS server IP address.

6. Test Using dig

Use dig for more detailed DNS information:

```
dig musicmania.in
```

To query a specific DNS server:

```
dig @192.168.1.10 musicmania.in
```

Check the **ANSWER SECTION** to confirm that the expected IP address is returned.

7. Check DNS Cache

If the DNS server has cached an incorrect or old record, clear the cache:

```
sudo rndc flush
```

Then test again:

```
dig @192.168.1.10 musicmania.in
```

Troubleshooting Checks Used

1. ping 8.8.8.8 — checks network connectivity.
2. named-checkconf — checks DNS configuration syntax.
3. named-checkzone — validates the musicmania.in zone file.
4. systemctl status bind9 — checks whether the DNS service is running.
5. nslookup musicmania.in — tests DNS resolution.
6. dig @DNS_SERVER musicmania.in — provides detailed DNS query results.
7. rndc flush — clears the DNS cache.